Title of the Project

Iris Flower Species Prediction Using Machine Learning

Description of the Project

The Iris Flower Prediction project is a machine learning–based web application developed to predict the species of an Iris flower. The system classifies the flower into one of three species: **Setosa, Versicolor, or Virginica**, based on four physical measurements — sepal length, sepal width, petal length, and petal width.

The application uses a pre-trained machine learning model stored in a pickle file. Users enter the flower measurements through a simple web interface, and the model predicts the corresponding Iris species. This project demonstrates the practical use of supervised machine learning and model deployment using a web application framework.

Tools and Libraries Used

Tool / Library	Purpose
Python	Core programming language
Streamlit	Web application framework
NumPy	Numerical data handling
Pickle	Loading trained ML model
Scikit-learn	Model training and prediction
VS Code	Development environment

Code Implementation (with Explanation)

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1. Importing Required Libraries

```
import streamlit as st
```

```
import pickle

import numpy as np
```

Explanation:

- streamlit is used to create the interactive web interface.
- pickle is used to load the trained machine learning model.
- numpy is used to handle numerical input values.

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2. Loading the Trained Model

```
model = pickle.load(open("iris.pkl", "rb"))
```

Explanation:
The trained Iris classification model is loaded from a pickle file. This avoids retraining the model each time the application runs.

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3. User Interface Design

```
st.title("🌸 Iris Flower Prediction App")

st.write("Enter the flower measurements below to predict the Iris species.")
```

Explanation:
This displays the application title and instructions, guiding the user on how to use the system.

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4. Accepting User Input

```
sepal_length = st.number_input("Sepal Length (cm)", min_value=4.3, max_value=7.9, step=0.1)

sepal_width  = st.number_input("Sepal Width (cm)", min_value=2.0, max_value=4.4, step=0.1)

petal_length = st.number_input("Petal Length (cm)", min_value=1.0, max_value=6.9, step=0.1)

petal_width  = st.number_input("Petal Width (cm)", min_value=0.1, max_value=2.5, step=0.1)
```

Explanation:
The user enters flower measurements using numeric input fields. Value limits ensure that only realistic data is entered.

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5. Preparing Input Data for Prediction

features = np.array([[sepal_length, sepal_width, petal_length, petal_width]])

Explanation:
The input values are combined into a NumPy array and reshaped to match the input format required by the machine learning model.

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6. Model Prediction

prediction = model.predict(features)[0]

Explanation:
The trained model predicts the class label corresponding to the Iris species.

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7. Displaying the Prediction Result

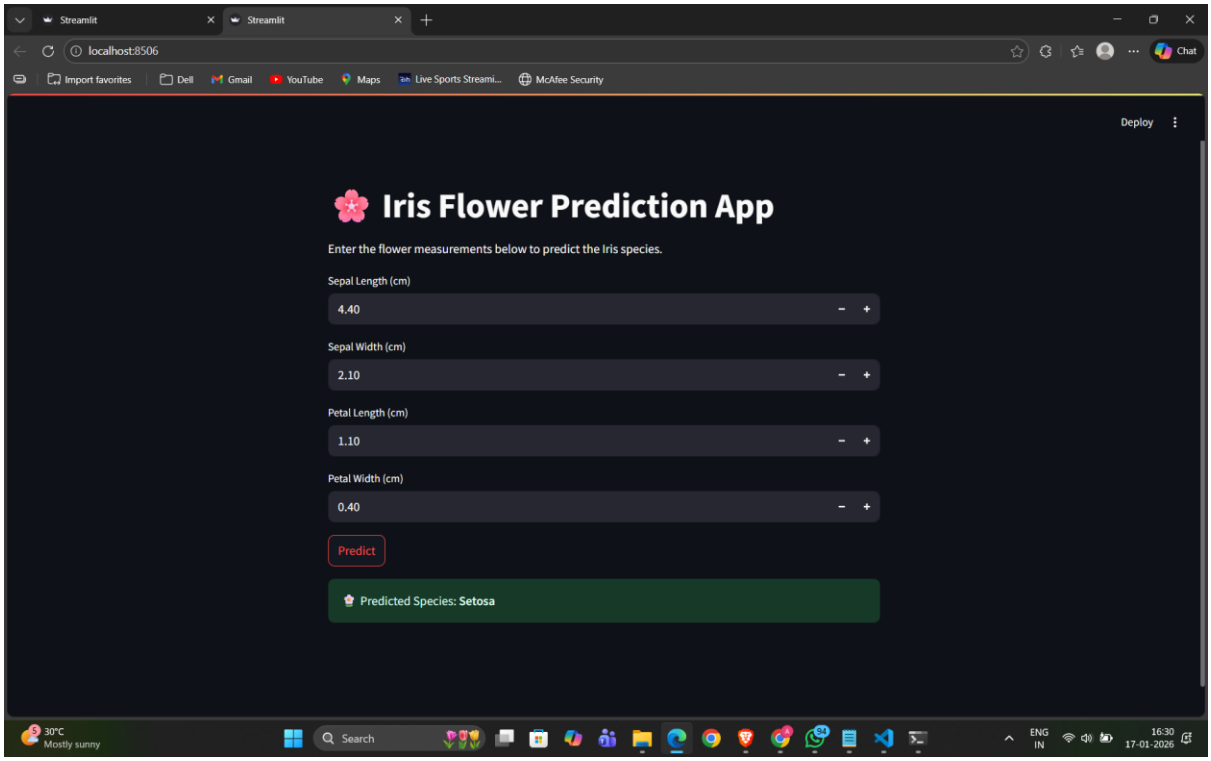
species_dict = {0: "Setosa", 1: "Versicolor", 2: "Virginica"}

st.success(f"🌼 Predicted Species: **{species_dict[prediction]}**")

Explanation:
The predicted class label is mapped to the actual flower species name and displayed clearly to the user.

Expected Output

- User enters flower measurements
- User clicks the Predict button
- The system processes the input using a trained ML model
- The predicted Iris species is displayed as:
    - Setosa, Versicolor, or Virginica
- Output is shown clearly on the web interface


Streamlit
Streamlit
localhost:8506
Import favorites
Dell
Gmail
YouTube
Maps
Live Sports Streami...
McAfee Security
Chat
Deploy
Iris Flower Prediction App
Enter the flower measurements below to predict the Iris species.
Sepal Length (cm)
4.40
Sepal Width (cm)
2.10
Petal Length (cm)
1.10
Petal Width (cm)
0.40
Predict
Predicted Species: Setosa
30°C
Mostly sunny
Search
ENG
IN
16:30
17-01-2026